Mathematical and Computational Methods

Exercise Sheet 5: Matrices

This sheet accompanies *Unit 5 (Matrices)*. It exercises matrix algebra and multiplication, the determinant and inverse, and the solution of linear systems; it matches the notation of Unit 5 and needs nothing beyond Units 1–5.

How to use this sheet. Problems are graded by difficulty—$\star$ drill (routine fluency), $\star\star$ core (standard, tutorial level) and $\star\star\star$ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit5}"`) and recompile.

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1 Warm-up drills

Short routine problems, at least one per skill. Keep the arithmetic tidy and check each answer.

Matrix algebra

Exercise 1.1 *($\star$ drill)***.** Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 4 \\ -2 & 5 \end{pmatrix}$. Compute $\mathbf{A}+\mathbf{B}$, $\mathbf{A}-\mathbf{B}$, $3\mathbf{A}$ and $2\mathbf{A}-\mathbf{B}$.

Exercise 1.2 *($\star$ drill)***.** Matrices have sizes $\mathbf{A}\colon 2\times 3$, $\mathbf{B}\colon 3\times 3$, $\mathbf{C}\colon 3\times 1$, $\mathbf{D}\colon 1\times 2$. For each product $\mathbf{AB}$, $\mathbf{BA}$, $\mathbf{BC}$, $\mathbf{CD}$, $\mathbf{DA}$, $\mathbf{AC}$, state whether it is defined and, if so, its size.

Exercise 1.3 *($\star$ drill)***.** Compute the product $\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 & 4 \\ 2 & -2 \end{pmatrix}$.

Exercise 1.4 *($\star$ drill)***.** For $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}$ write down $\mathbf{A}^\mathsf{T}$ and compute $\operatorname{tr}(\mathbf{A})$.

Determinants and inverses

Exercise 1.5 *($\star$ drill)***.** Evaluate the three determinants

$$\begin{vmatrix} 3 & 5 \\ 1 & 2 \end{vmatrix}, \qquad \begin{vmatrix} 4 & 2 \\ 6 & 3 \end{vmatrix}, \qquad \begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix}.$$

Exercise 1.6 *($\star$ drill)***.** Find the inverse of $\mathbf{M} = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$ and verify $\mathbf{MM}^{-1} = \mathbf{I}$.

Exercise 1.7 *($\star$ drill)***.** Evaluate $\det\begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & -1 \\ 0 & 2 & 4 \end{pmatrix}$ by cofactor expansion along the first row.

Linear systems

Exercise 1.8 *(⋆ drill)*. Solve by the inverse method: $3x + 5y = 1,\ x + 2y = 1$. (You may reuse $\mathbf{M}^{-1}$ from Exercise 1.6.)

Exercise 1.9 *(⋆ drill)*. Solve by elimination (write the augmented matrix and reduce): $x + y = 5,\ 2x - y = 1$.

Transformations

Exercise 1.10 *(⋆ drill)*. Write down the 2×2 matrices for (a) a rotation by $90°$ anticlockwise, (b) reflection in the x-axis, (c) an enlargement by factor 3. Then apply the rotation to $\mathbf{v} = (2, 1)$.

2 Tutorial problems

The core set for the 2-hour class. These are anchored on the worked examples of Unit 5 and spread applied context across the three cohorts. Standard difficulty unless marked. Problems flagged ▶ are the live tutorial core to work through in class; the remaining problems in this section round out the topic and are for completion at home.

Exercise 2.1 *(▶ ⋆⋆ core)*. Let $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$.

(a) Compute $\mathbf{AB}$ and $\mathbf{BA}$ and confirm they differ.

(b) Verify $(\mathbf{AB})^\mathsf{T} = \mathbf{B}^\mathsf{T}\mathbf{A}^\mathsf{T}$ for these matrices.

Exercise 2.2 *(▶ ⋆⋆ core— data science)*. A shop records units sold of three products on two days as the rows of $\mathbf{S} = \begin{pmatrix} 4 & 2 & 1 \\ 0 & 3 & 5 \end{pmatrix}$ (rows = days, columns = products). The unit prices are $\mathbf{p} = (3, 5, 2)^\mathsf{T}$.

(a) Explain why $\mathbf{Sp}$ is defined and what its entries mean.

(b) Compute the daily revenue vector $\mathbf{Sp}$.

Exercise 2.3 *(⋆⋆ core)*. Let $\mathbf{T} = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}$, whose columns are the vectors $\mathbf{u} = (1, 2, 0)$, $\mathbf{v} = (0, 1, 1)$, $\mathbf{w} = (1, 0, 2)$.

(a) Compute $\det \mathbf{T}$ by cofactor expansion.

(b) The volume of the parallelepiped on $\mathbf{u}, \mathbf{v}, \mathbf{w}$ is the scalar triple product $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$ (Unit 4). Confirm it equals $|\det \mathbf{T}|$.

Exercise 2.4 *(▶ ⋆⋆ core— the threaded system, part 1)*. Let $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$.

(a) Show $\det \mathbf{A} = 1$.

(b) Find $\mathbf{A}^{-1}$ by the cofactor (adjugate) method and verify $\mathbf{AA}^{-1} = \mathbf{I}$.

We reuse this matrix in Exercise 2.5.

Exercise 2.5 (▶ ⋆⋆ *core— the threaded system, part 2: two roads*)**.** Solve $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A}$ from Exercise 2.4 and $\mathbf{b} = (6, 5, 11)^{\top}$, that is

$$\begin{aligned} x + 2y + 3z &= 6, \\ y + 4z &= 5, \\ 5x + 6y \quad\quad &= 11, \end{aligned}$$

twice: (a) by the inverse method, and (b) by Gaussian elimination on the augmented matrix. Confirm the two answers agree.

Exercise 2.6 (▶ ⋆⋆ *core— infinitely many solutions*)**.** Solve, by elimination,

$$\begin{aligned} x + 2y - z &= 3, \\ 2x + y + z &= 3, \\ x - y + 2z &= 0, \end{aligned}$$

and write the solution set as the vector equation of a line (Unit 4). How does the echelon form reveal that the system has infinitely many solutions?

Exercise 2.7 (⋆⋆ *core— spot the case*)**.** Apply elimination to

$$x + y + z = 1, \qquad x + 2y + 3z = 2, \qquad 2x + 3y + 4z = 5,$$

and explain what the echelon form tells you about the solution set.

Exercise 2.8 (▶ ⋆⋆ *core— physics*)**.** A force $\mathbf{F} = (2, 3)$ (in newtons) is first rotated 90° anticlockwise and then reflected in the x-axis.

(a) Find the single 2×2 matrix $\mathbf{M}$ representing "rotate, then reflect".

(b) Compute the transformed force $\mathbf{MF}$.

(c) What single geometric transformation is $\mathbf{M}$?

Exercise 2.9 (⋆⋆ *core*)**.** For each 2×2 transformation matrix below, compute the determinant and state its effect on area and orientation, and whether the map is invertible:

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{N} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Exercise 2.10 (▶ ⋆⋆ *core— conceptual, true or false*)**.** Decide whether each statement is true or false; give a one-line reason or counterexample.

(a) Matrix multiplication is commutative.

(b) Every square matrix has an inverse.

(c) $(\mathbf{AB})^{\top} = \mathbf{A}^{\top}\mathbf{B}^{\top}$.

(d) If $\det \mathbf{A} = 0$ then $\mathbf{Ax} = \mathbf{b}$ cannot have a unique solution.

(e) $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$.

Exercise 2.11 (⋆⋆ *core— maths*)**.** Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. Prove directly that $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$.

Exercise 2.12 *(⋆⋆ core— data science: Markov chains (foreshadows Unit 6))***.** A simple weather model has two states, *sunny* (S) and *rainy* (R). The *transition matrix* (columns = today, rows = tomorrow)

$$\mathbf{P} = \begin{pmatrix} 0.8 & 0.4 \\ 0.2 & 0.6 \end{pmatrix}$$

reads: if today is sunny, tomorrow is sunny with probability 0.8 and rainy with 0.2 (top and bottom of the first column), and similarly the second column handles a rainy today. Today is sunny, so the state vector is $\mathbf{x}_0 = (1, 0)^\top$.

(a) Compute $\mathbf{x}_1 = \mathbf{P}\mathbf{x}_0$ and say what its entries mean.

(b) Compute $\mathbf{P}^2$, then $\mathbf{x}_2 = \mathbf{P}^2\mathbf{x}_0$. Interpret the $(1, 1)$ entry of $\mathbf{P}^2$ (the probability of being sunny two days after a sunny day).

3 Further practice (home)

A larger bank for independent study, running from routine to harder, ending with a few challenge/extension problems marked ⋆ ⋆ ⋆.

Matrix algebra

Exercise 3.1 *(⋆⋆ core)***.** Compute $\mathbf{AB}$ where $\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 0 \end{pmatrix}$. State the size of the result first.

Exercise 3.2 *(⋆⋆ core)***.** Verify the "inverse of the transpose" rule $(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$ for $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$. (This is a genuinely different identity from the product–transpose rule $(\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top$ that you already meet in Exercises 2.1 and 2.10.)

Exercise 3.3 *(⋆⋆ core)***.** Let $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Compute $\mathbf{A}^2$ and $\mathbf{A}^3$, then conjecture a formula for $\mathbf{A}^n$.

Exercise 3.4 *(⋆⋆ core)***.** Show that for any square matrix $\mathbf{A}$ the matrix $\mathbf{A} + \mathbf{A}^\top$ is symmetric, and illustrate with $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$.

Determinants and inverses

Exercise 3.5 *(⋆⋆ core)***.** Find the inverse of each, or state that none exists:

$$\text{(a)} \begin{pmatrix} 4 & 7 \\ 1 & 2 \end{pmatrix}, \qquad \text{(b)} \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}, \qquad \text{(c)} \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}.$$

Exercise 3.6 *(⋆⋆ core)***.** Evaluate $\det \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 0 \\ 5 & 6 & 1 \end{pmatrix}$, choosing a row or column to minimise work.

Exercise 3.7 *(⋆⋆ core)***.** Find the inverse of $\mathbf{U} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ and verify $\mathbf{UU}^{-1} = \mathbf{I}$.

Exercise 3.8 *($\star\star$ core)*. For which value(s) of k is each matrix singular?

$$\text{(a)}\ \begin{pmatrix} 1 & k \\ k & 4 \end{pmatrix}, \qquad \text{(b)}\ \begin{pmatrix} 1 & 2 & 3 \\ 0 & k & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Linear systems

Exercise 3.9 *($\star\star$ core)*. Solve by the inverse method, reusing $\mathbf{U}^{-1}$ from Exercise 3.7:

$$x + y + z = 4, \qquad y + z = 2, \qquad z = 1.$$

Exercise 3.10 *($\star\star$ core)*. Solve by Gaussian elimination:

$$x + y + z = 6, \qquad 2x - y + z = 3, \qquad x + 2y - z = 2.$$

Exercise 3.11 *($\star\star$ core— Gauss–Jordan)*. Take the row-echelon form reached in Exercise 3.10,

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array}\right),$$

and continue the row operations to *reduced* row-echelon form (leading 1s, with zeros both below *and* above each pivot). Read the solution off directly, without any back-substitution.

Exercise 3.12 *($\star\star$ core— conceptual)*. Each augmented matrix below is already in echelon form. Classify the system as having a unique solution, no solution, or infinitely many.

$$\text{(a)}\ \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 4 \end{array}\right), \quad \text{(b)}\ \left(\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 \end{array}\right), \quad \text{(c)}\ \left(\begin{array}{ccc|c} 1 & 0 & -2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Exercise 3.13 *($\star\star$ core)*. Find all solutions of $x + y + z = 3,\ x - y + z = 1$ and write the answer as the vector equation of a line.

Exercise 3.14 *($\star\star$ core— physics)*. Mesh analysis of a two-loop circuit gives the currents I_1, I_2 (in amperes) from

$$5I_1 - 2I_2 = 1, \qquad -2I_1 + 4I_2 = 6.$$

Solve by the inverse method.

Transformations

Exercise 3.15 *($\star\star$ core)*. Write the matrix for reflection in the line $y = x$, and use it to find the image of $(3, 5)$. Write also the matrices for reflection in the y-axis and for a $180°$ rotation.

Exercise 3.16 *($\star\star$ core)*. Let $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ (a horizontal stretch) and $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (rotation $90°$). Find the matrix for "stretch first, then rotate", and its determinant. Does the order matter?

Exercise 3.17 *($\star\star$ core— sketch it)*. Let $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ (a horizontal stretch) and $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (rotation by $90°$). On graph paper, sketch the unit square with corners $(0,0), (1,0), (1,1), (0,1)$, then sketch its image under $\mathbf{D}$ and its image under $\mathbf{RD}$ (stretch first, then rotate). In each case find the area of the image and check that it equals $|\det|$ times the original area.

Challenge / extension

Exercise 3.18 *($\star\star\star$ challenge)*. Suppose $\mathbf{A}$ and $\mathbf{B}$ are invertible $n \times n$ matrices. Prove that $\mathbf{AB}$ is invertible with $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

Exercise 3.19 *($\star\star\star$ challenge— links to Unit 3)*. Let $\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. By computing $\mathbf{R}(\theta)\mathbf{R}(\phi)$, show that $\mathbf{R}(\theta)\mathbf{R}(\phi) = \mathbf{R}(\theta + \phi)$, and read off the angle-addition formulae for cos and sin.

Exercise 3.20 *($\star\star\star$ challenge)*. Find all 2×2 matrices $\mathbf{X}$ that commute with $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, i.e. $\mathbf{XJ} = \mathbf{JX}$.

Exercise 3.21 *($\star\star\star$ challenge— spot the error)*. A student "inverts" $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$ by writing $\mathbf{A}^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/4 \end{pmatrix}$ "because you invert each entry," and then claims the same recipe gives $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 1/2 \\ 1/3 & 1/4 \end{pmatrix}$. Identify the error and give the correct inverse of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Exercise 3.22 *($\star\star\star$ challenge)*. Suppose a square matrix satisfies $\mathbf{A}^2 = \mathbf{A}$ (it is *idempotent*) and is also invertible. Prove that $\mathbf{A} = \mathbf{I}$.